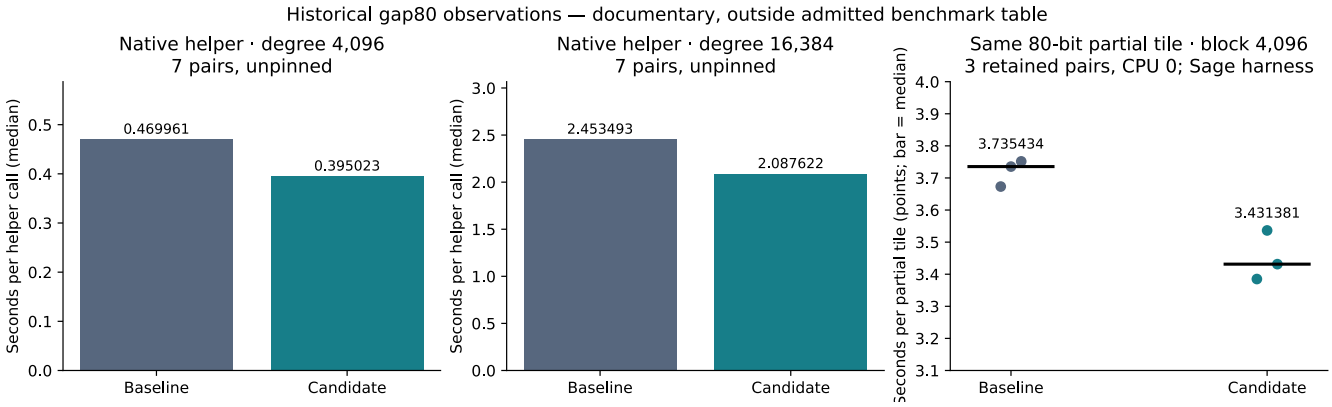
Gap80 frozen run record

Documentary publication, 31 October 2026. Source: the unchanged `../evidence.tar.gz` package and `../historical-summary.json`. No new timing experiment was launched while preparing this PR.



Workload, arms and accounting

The native helper computes three log-norm coefficients. Its frozen baseline uses three trace-only polynomial products; the candidate shares a trace functional and reversed-modulus inverse. Deterministic random-polynomial microbenchmarks used FLINT seeds 20261031 and 80, seven alternating-order pairs at each degree (4,096 and 16,384). Both arms include text conversion, modulus checking and polynomial work. They measure the helper only, not construction of an isogeny. The exact inputs are generated by archived `bench.c`; each pair's outputs were checked identical.

The separate tile workload fixes the actual 80-bit kernel, block 4,096 and three contexts (infinity, G and 7G). It measures one partial tile with the Sage/Python harness and native helper. It excludes process startup and is not a complete quotient or point-map measurement. All seven tile pairs have equal partial-jet outputs.

Recorded flow and exclusions

Helper: 14 paired repeats completed, seven per polynomial degree, all retained; unpinned shared-host diagnostics. Tile: seven paired repeats completed; pairs 1–3 unpinned, with pair 1 overlapping the fixture selftest; pairs 4–7 pinned to CPU 0, but pair 4 overlapped the root-oracle check. Only pairs 5–7 enter the tile summary. This rule and every receipt are retained in the frozen package. No unsuccessful performance run is hidden.

Baseline/candidate tile medians were 3.735434/3.431381 s; the three retained observations span 3.673185–3.751641 s and 3.384988–3.536417 s respectively. No confidence interval, A/A spread, host-quiescence certificate, fastest-tool comparison or independent timing replay was recorded. Pinning is not a claim of exclusive host access. Do not promote these clocks as a current native end-to-end or general wall-time advantage.

Environment, checks and failures

AMD EPYC 9V74, Linux x86-64, FLINT 3.3.1; exact compiler/platform are in the archive's `environment.json`. Native direct-root checks: 96 cases and 96 noninvertibility controls over three fields. Historical complete Sage fixtures: degrees 19 and 10,009, six point images each, agreement with traditional Vélu. Actual target: one 128-root tile at 32 non-kernel scalar contexts, three backends, independently enumerated roots. These are partial sums, not 32 full target-size map evaluations.

Historical setup failures retained in README/provenance: missing benchmark headers (corrected), wrong initial curve twist (both tested; the +784 model passes independent verification), read-only Sage cache (explicit scratch cache used), and unavailable external timing wrapper (native resource capture used). Publication-time Cargo is unavailable (exit 127); native build, direct-root and PARI correctness checks pass. No new Rust path changes.

Reproducibility and sources

Extract the archive and run its SHA256SUMS to verify all 55 frozen files. The native-only entry points and exact fixed source/kernel parameters are in `../native/` and `../parameters.json`. The package hash is `630e66aa7101ad0be6290e9e26acc969925e9ef07c488df20488370fee4a1e44`. The study README links FLINT, Sage 10.9 square-root Vélu and PARI primary documentation and the related gap60 archive. No new performance admission or largest-known status is asserted by this report.